

TO - 92 BIPOLAR TRANSISTORS
TRANSISTOR(NPN)

FEATURES

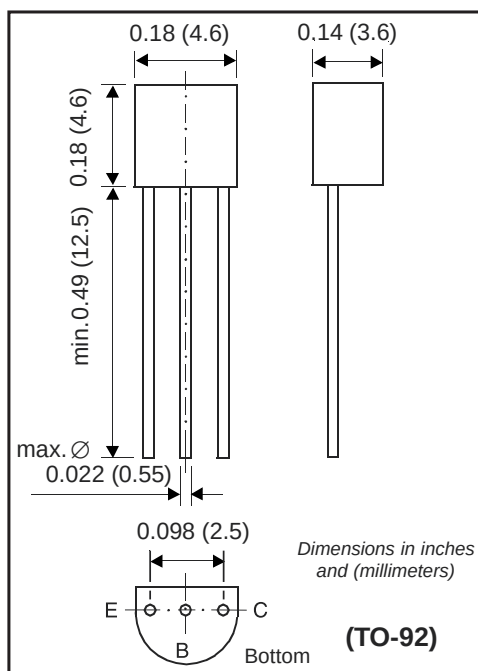
- * Power dissipation
P_{CM}: 0.63 W(T_{amb}=25°C)
- * Collector current
I_{CM}: 0.2 A
- * Collector-base voltage
V_{(BR)CBO}: 60 V
- * Operating and storage junction temperature range
T_J, T_{stg}: -55°C to +150°C

MECHANICAL DATA

- * Case: Molded plastic
- * Epoxy: UL 94V-O rate flame retardant
- * Lead: MIL-STD-202E method 208C guaranteed
- * Mounting position: Any
- * Weight: 0.008 gram

MAXIMUM RATINGS AND ELECTRICAL CHARACTERISTICS

Ratings at 25°C ambient temperature unless otherwise specified.
Single phase, half wave, 60 Hz, resistive or inductive load.
For capacitive load, derate current by 20%.



MAXIMUM RATINGS (@ T_A = 25°C unless otherwise noted)

RATINGS	SYMBOL	VALUE	UNITS
Max. Steady State Power Dissipation ⁽¹⁾ @T _A =25°C Derate above 25°C	P _D	625	mW
Max. Operating Temperature Range	T _J	150	°C
Storage Temperature Range	T _{STG}	-55 to +150	°C

ELECTRICAL CHARACTERISTICS (@ T_A = 25°C unless otherwise noted)

CHARACTERISTICS	SYMBOL	MIN.	TYP.	MAX.	UNITS
Thermal Resistance Junction to Ambient	R _{θJA}	-	-	200	°C/W

Notes : 1. Alumina=0.4*0.3*0.024in.99.5% alumina
2. "Fully ROHS Compliant", "100% Sn plating (Pb-free)".

ABSOLUTE MAXIMUM RATINGS (T_a=25°C)

DESCRIPTION	SYMBOL	VALUE	UNITS
Collector Emitter Voltage	V _{CEO}	40	V
Collector Base Voltage	V _{CBO}	60	V
Emitter Base Voltage	V _{EBO}	6.0	V
Collector Current Continuous	I _C	200	mA
Power Dissipation at T _a =25°C	P _D	625	mW
Derate Above 25°C		5.0	mW/°C
Power Dissipation at T _c =25°C	P _D	1.5	W
Derate Above 25°C		12	mW/°C
Operating and Storage Junction Temperature Range	T _j , T _{stg}	-55 to +150	°C

THERMAL RESISTANCE

Junction to Case	R _{th (j-c)}	83.3	°C/W
Junction to Ambient in free air	R _{th (j-a)}	200	°C/W

ELECTRICAL CHARACTERISTICS (T_a=25°C unless specified otherwise)

DESCRIPTION	SYMBOL	TEST CONDITION	2N3903	2N3904	UNITS
Collector Emitter Voltage	V _{CEO}	I _C =1mA, I _B =0	>40	>40	V
Collector Base Voltage	V _{CBO}	I _C =10μA, I _E =0	>60	>60	V
Emitter Base Voltage	V _{EBO}	I _E =10μA, I _C =0	>6.0	>6.0	V
Base Cut Off Current	I _{BL}	V _{CE} =30V, V _{EB} =3V	< 50	< 50	nA
Collector Cut Off Current	I _{CEX}	V _{CE} =30V, V _{EB} =3V	< 50	< 50	nA
DC Current Gain	*h _{FE}	I _C =0.1mA, V _{CE} =1V I _C =1mA, V _{CE} =1V I _C =10mA, V _{CE} =1V I _C =50mA, V _{CE} =1V I _C =100mA, V _{CE} =1V	>20 >35 50-150 >30 >15	>40 >70 100-300 >60 >30	
Collector Emitter Saturation Voltage	*V _{CE (sat)}	I _C =10mA, I _B =1mA I _C =50mA, I _B =5mA	< 0.2 < 0.3	< 0.2 < 0.3	V V
Base Emitter Saturation Voltage	*V _{BE (sat)}	I _C =10mA, I _B =1mA I _C =50mA, I _B =5mA	0.65 - 0.85 < 0.95	0.65 - 0.85 < 0.95	V V

*Pulse Condition: =300ms, Duty Cycle=2%

SMALL SIGNAL CHARACTERISTICS

DESCRIPTION	SYMBOL	TEST CONDITION	2N3903	2N3904	UNITS
Transistors Frequency	f _T	I _C =10mA, V _{CE} =20V, f=100MHz	>250	>300	MHz
Output Capacitance	C _{ob}	V _{CB} =5V, I _E =0, f=1MHz	< 4.0	<4.0	pF
Input Capacitance	C _{ib}	V _{EB} =0.5V, I _C =0, f=1MHz	< 8.0	< 8.0	pF
		ALL f=1kHz			
Small Signal Current Gain	h _{fe}	I _C =1mA, V _{CE} =10V	50 - 200	100 - 400	
Input Inpedence	h _{ie}	I _C =1mA, V _{CE} =10V	1.0 - 8.0	1.0 -10	kΩ
Voltage Feedback Ratio	h _{re}	I _C =1mA, V _{CE} =10V	0.1 - 5.0	0.5 - 8	x10 ⁻⁴
Out put Adimttance	h _{oe}	I _C =1mA, V _{CE} =10V	1.0 - 40	1.0 - 40	μmhos
Noise Figure	NF	I _C =100μA, V _{CE} =5V, f=1KHz, R _S =1KΩ	< 6.0	< 5.0	dB
SWITCHING Time					
Delay time	t _d	V _{CC} =3V, V _{BE} =0.5V	< 35	< 35	ns
Rise time	t _r	I _C =10mA, I _{B1} =1mA	< 35	< 35	ns
Storage time	t _s	V _{CC} =3V, I _C =10mA	< 175	< 200	ns
Fall time	t _f	I _{B1} =1B ₂ =1mA	< 50	< 50	ns

DISCLAIMER NOTICE

Rectron Inc reserves the right to make changes without notice to any product specification herein, to make corrections, modifications, enhancements or other changes. Rectron Inc or anyone on its behalf assumes no responsibility or liability for any errors or inaccuracies. Data sheet specifications and its information contained are intended to provide a product description only. "Typical" parameters which may be included on RECTRON data sheets and/ or specifications can and do vary in different applications and actual performance may vary over time. Rectron Inc does not assume any liability arising out of the application or use of any product or circuit.

Rectron products are not designed, intended or authorized for use in medical, life-saving implant or other applications intended for life-sustaining or other related applications where a failure or malfunction of component or circuitry may directly or indirectly cause injury or threaten a life without expressed written approval of Rectron Inc. Customers using or selling Rectron components for use in such applications do so at their own risk and shall agree to fully indemnify Rectron Inc and its subsidiaries harmless against all claims, damages and expenditures.